3
Major Sound Changes

The basic meaningful elements, or morphemes, that comprise Skiri words generally have a single basic (or underlying) form. However, when they are combined with other morphemes to form words that are actually spoken, sound changes occur that fuse those morphemes into fully formed words. Often the changes are small, altering a consonant or vowel of a morpheme, but just as frequently the changes can be more extensive, sometimes resulting in contractions that mask the underlying identities of morpheme combinations. The user must be aware of these sound changes in order to understand word formation in Skiri and, more importantly, to create new words from the information provided in dictionary entries.

Presented here is a summary of the major Skiri phonological processes that occur when combining morphemes to create new words. There are two types of processes, or sound rules: **general rules** that apply to all instances of specified sound sequences, no matter what the constituent morphemes are; and **restricted rules** that apply only to sequences of specified morphemes, which are highly restricted in number

Vowels: Restricted Rules

There are five restricted rules that apply to vowel sequences in specific morpheme sequences.

Rule 1R. Dominant *i*. There are seven morpheme combinations in which a final *i* or *ii* in the preceding morpheme and the initial *u* of the following morpheme contract to *i* or *ii*. Those combinations are the following:

Preceding Prefixes	Following Prefix or Stem-Initial Vowel	Resulting Contraction
rii- 'assertive mode (ASSR)'	*-uks-* 'aorist (AOR)'	*riiks-*
[*ti-* 'indicative mode (IND.3.A)' *ii-* 'subjunctive mode (SUBJ.3.A)' *ri-* 'contingent mode (CONT.3.A)']	*u* . . .]$_{\text{STEM}}$	[*tii-* *ii-* *rii-*]
[*-i-* 'sequential (SEQ)' *-ir-* 'preverb (PREV)' *acir-* 'inclusive dual agent (INCL.DU.A)']	*-uur-* 'preverb (PREV)'	[*-iir-* *-iir-* *-aciir-*]

Thus, the restricted rule:

i, ii + *u* → *i, ii*

Examples:

tiihura 'he is alone' (< *ti-* IND + Ø- 3.A + *uhurah* 'be alone' + -Ø PERF)

taciihii 'you and I are good' (< *ta-* IND.1/2.A + *aciir-* 1.DU.IN.A + *uur-* PREV + *hiir* 'be good' + -Ø PERF)

Rule 2R. Dominant *a*. Although the vowel sequence *i* + *a* contracts to *ii*, in certain morpheme sequences the combination results in *a*; thus,

i + *a* → *a*

This rule occurs when any of the following modal prefixes is the first element and one of the *a*-dominant morphemes is the second element in the contraction:

Modal Prefixes		***a*-Dominant Morphemes**	
ti-	indicative mode, third person agent (IND)	*a-*	second person patient (2.P)
ri-	contingent mode, third person agent (CONT)	*a-*	preverb (PREV); third person agt. possessor (3.POSS.A)
kuus...i-	potential mode, third person agent (POT.1)	*aca-*	inclusive dual patient (1D.IN.P)
ii-	subjunctive mode, third person agent (SUBJ)	*ar-*	evidential proper (EV)

Examples of the preceding combinations are the following (with contracting vowels in boldface):

*t**a**piruʔ* 'he whipped you' < ***ti-*** **IND** + Ø- 3.A + ***a-*** **2.P** + *piru* 'whip' + -Ø PERF

kusaa**kiraawaʔ* 'you will forget' < ***kuus- **POT.1/3A** + ***a-*** **2.P** + ***i-*** **SEQ** + *kiraawaahc* 'be sick' + -Ø PERF

*wiit**ar**uutasitit* 'it would happen' < *wii-* 'now' + ***ti-*** **IND** + Ø- 3.A + ***ar-*** **EV** + *ut-* PREV + *asitik* 'happen' + -Ø PERF

*si**ʔaawa**ʔa* 'they (DU) should have come' < *si-* DU + ***ii-*** SUBJ + Ø- 3.A + *a-* PREV + *wa-* DIST + *a* 'come' + -Ø SUB.3

Rule 3R. *-his* 'perfective' Reduction. The suffix *-his* 'perfective (PERF)' reduces to *s* after verb stems with a final *k*; thus,

$Vk]_{\text{STEM}}$ + *-his* → *ks*

An example of this rule is:

> *haks-* 'pass by (PERF)' < ***hak* 'pass by' + *-his* PERF**, as in *tihaksta* 'he is going to pass by' < *ti-* IND + *Ø-* 3.A + *hak* 'pass by' + *-his* PERF + *-ta* INT

Rule 4R. Stem-final Vocalic Reduplication. This rule, which applies to a class of descriptive verbs—approximately half of all descriptives—in their non-subordinate forms is

$$V_i(V_i) \rightarrow V_i(V_i)ʔV_i \ / \ C_C + Ø \ (+ \text{NON-SUB})$$

> *tipahaaʔat* 'it is red' < *ti-* IND + *Ø-* 3.A + *pahaat* 'be red' + *Ø* 'be'; cf. *iriirapahaatu* 'the one that is red'

> *tihuraahtarariʔit* 'the ground is burned off' < *ti-* IND + *Ø-* 3.A + *huraar-* 'ground' + *tararit* 'be burned, scorched' + *-Ø* 'be'; cf. *iriiratararitu* 'where it is scorched'

Vowels: Unrestricted Rules

Two or more contiguous vowels, either across a morpheme boundary or within a morpheme, contract, except when either of two conditions occurs:

(1) the second of two vowels is long, or
(2) two vowels are in word-final position or preceding a final consonant.

Vowel contraction takes place when there is a sequence of a short or long vowel followed by a short vowel before one or two consonants. Thus,

$$V, V_iV_i + V \rightarrow VV \ / \ _ \ C(C)V$$

(where V_i represents a vowel that is identical to another so-marked vowel in a rule and V_iV_i represents a long vowel).

Rule 5. Same-vowel Contractions. When two identical vowels are contiguous, they contract to an identical long vowel. The first vowel in the sequence may be short or long. Thus,

$$V_i(V_i) + V_i \rightarrow V_iV_i$$

Rule 6. Contractions of Any Vowel + *u*: *u*-Domination. In a sequence of two vowels, in which the first one may be long or short, and where any vowel in the sequence is *u*, the vowels contract to a long *uu*.

$$\begin{bmatrix} V, V_iV_i + u \\ u(u) + V \end{bmatrix} \rightarrow uu \ / \ _ \ C(C)V$$

Examples are:

> *tuuriciisat* 'he is tired' < *ti-* IND + Ø- 3.A + Ø- 3.P + *uur-* PREV + *iciisat* 'be tired' + -Ø PERF; *tuuriciisat* 'you are tired' < *ti-* IND + Ø- 3.A + *a-* 2.P + *uur-* PREV + *iciisat* 'be tired' + -Ø PERF
>
> *kuʔuutkaksaa* 'if he could holler for him' < *ku-* INDF + *ii-* SUBJ + Ø- 3.A + *ut-* BEN + *kaksaa* 'holler, yell' + -Ø SUB.4

Rule 7. Contractions of *i* and *a*. In combinations of *i* or *ii* followed by *a* or a combination of *a* or *aa* followed by *i*, the vowels contract to *ii*. (The result of contractions of *i* and *a* in South Band Pawnee and Arikara is *ee*.)

$$\begin{bmatrix} i(i) & + & a \\ a(a) & + & i \end{bmatrix} \rightarrow ii \ / \ _ \ \text{C(C)V}$$

For example,

> *tiitkuʔ* 'he hears' < *ti-* IND + Ø- 3.A + *atkaʔu* 'hear' + -Ø PERF

Consonants: Restricted Rules

There are four commonly occurring restricted rules for consonants. Those rules are the following.

Rule 8R. Assibilation of *raar-*. This rule applies to the final consonant of *raar-* 'plural; iterative' when followed by a stem-initial *t*. The *r* + *t* sequence results in a contraction to *c*, as follows:

$$r + t \rightarrow c \ / \ raar\text{-} + t \ldots]_{\text{STEM}}$$

Example:

> *tira**c**akasis* 'they are close together' < *ti-* IND + Ø- 3.A+ *raa**r**-* PL.INDV.A + ***t**akasis* 'be together' + -Ø PERF

Rule 9R. Final-*s* Loss Rule. Certain nouns and suffixes that end in *s* regularly lose that *s* when it is in word-final position, but retain it when another morpheme follows the stem in derivation or inflection. The rule is:

$$s \rightarrow \emptyset \ / \ _ \ \#$$

A noun illustrating final-*s* loss is the form *kuhkus-* 'swine, pig; pork', which in independent form is always *kuhku* but is *kuhkus* in compounds, as in *kuhkuskaʔit* 'salt pork' and *kuhkuskaasuʔ* 'ham'. A common example of this rule is the aspectual suffix *-:hus/-hus* 'imperfective (IMPF)' that loses its final *s* when the suffix is word final; for example, *tikiikaahuʔ* 'he is drinking' (< *ti-* IND + Ø- 3.A + *kiikaa* 'drink' + *-:hus* IMPF).

Rule 10R. Prefixal Final-*r* Loss. Two prefixes, *ir-* 'preverb (PREV); third person transitive agent plural (PL.3.A); obviative (OBV)' and *acir-* 'first person inclusive dual agent (1D.IN.A)' lose their final *r* when they contract with any of the three prefixes, *uur-* 'preverb (PREV); patient possessor (P.POSS)'; *ut-* 'preverb (PREV); dative/ benefactive (BEN)'; and *uks-* 'aorist (AOR); jussive (JUSS)'. When final *r* of *ir-* and *acir-* is lost, the resulting final *i* of the two forms becomes dominant in the resulting contractions with initial *u* of the preceding three morphemes. Example:

> *tat**iit**urikataku* 'I am carrying it under my arm' < *ta-* IND.1/2.A + *t-* 1.A + ***ir-*** **PREV** + ***ut-*** **PREV** + *urikatakus* 'carry under one's arm' + -Ø PERF

Rule 11R. *ut-* Affrication Rule. Final *t* of the prefix *ut-* 'preverb (PREV); dative/ benefactive (DAT/BEN)' affricates to *c* when followed by *i-* 'sequential (SEQ)'. For example,

> *kuustuciʔa* 'I will do it' < *kuus . . . i-* POT.1/3.A + *t-* 1.A + *ut-* PREV + *aar* 'do' + -Ø PERF; cp. *tatuuta* 'I did it' < *ta-* IND.1/2.A + *t-* 1.A + *ut-* PREV + *aar* 'do' + -Ø PERF

Thus,

$$ut\text{-} \rightarrow uc\text{-} \;/\; _\; i\text{-}$$

In Skiri *i-* SEQ does not dominate over a following *u* in *-uks* AOR; JUSS, as it does in South Band. For example,

> *sucuksa* 'do it!' < *i-* CONT.1/2.A + *s-* 2.A + *ut-* PREV + *i-* SEQ + *uks-* JUSS + *aar* 'do' + -Ø PERF; cp. South Band *suciksa* 'do it!'

Rule 12R. Prefixal *r*-Laryngealization Rule. This rule applies only to sequences of *r* + *r* in which the preceding *r* is the final sound of one of the following two evidential prefixes:

tiir-	'inferential (INFR)'
ar-	'evidential (proper) (EV)'

The final *r* of these prefixes changes to *h* when it precedes an *r*-initial morpheme. Thus,

$$r \rightarrow h \;/\; _\; r$$

Examples are:

> *a**h**iʔat* 'he went' < *ar-* EV + *ri-* CONT.3.A + Ø- 3.A + *at* 'go' + -Ø PERF

> *ti**ih**akuutit* 'he must have killed it' < *tiir-* INF + *ra-* ABS + Ø- 3.A + *kuutik* 'kill' + -Ø PERF

Note that this rule must be applied before the more general Consonantal Degemination Rule below, which reduces *r* + *r* sequences to a single *r*. It also must precede the *r*-Reduction Rule below, which deletes the *r* of an *h* + *r* sequence.

Consonants: Unrestricted Rules

The following rules apply generally to consonants when the specified conditions occur.

Rule 13. *t*-Laryngealization Rule. The stop *t* becomes laryngeal *h* when it precedes *r*. The rule can be symbolized thus:

$$t \rightarrow h \ / \ _ \ r$$

Examples are:

> *tahuukuʔ* 'I am making it' < *ta-* IND.1/2.A + *t-* 1.A+ *raʔuk* 'make' + *-:hus* IMPF
>
> *pahaahaʔuk* 'to redden' < *pahaat* 'be red' + *raʔuk* 'make; CAUS'

Rule 14. Metathesis Rule. When underlying *r* is followed by *h*, the two sounds metathesize; hence

$$r + h \rightarrow hr$$

However, the sequence *hr* never occurs in spoken (or surface) forms in Skiri (as it does in South Band) because an *hr* sequence always reduces to *h*, as in the following rule. Note that the three sequential rules here—*t*-Laryngealization, Metathesis, and Sonorant Reduction—must be applied in the order given; that is, *t*-Laryngealization must be applied first and Sonorant Reduction last. An example of this rule is:

> *tuuhii* 'he is good' < *ti-* IND + Ø- 3.A + *uur-* PREV + *hiir* 'be good' + -Ø PERF; to obtain *tuuhii*, the following Sonorant *r*-Reduction Rule must be applied.

Rule 15. Sonorant Reduction Rule. This rule reduces the cluster *hr* to *h*; hence

$$r \rightarrow \emptyset \ / \ _ \ h$$

> *tahahuukat* 'I took it inside' < *ta-* IND.1/2.A + *t-* 1.A + Ø- + *rahuukat* 'take inside' + -Ø PERF

Rule 16. *h*-Loss Rule. Laryngeal *h* is lost in several environments, one of which is when it is preceded by a consonant. Thus,

$$h \rightarrow \emptyset \ / \ C \ _$$

> *taturaas* 'I found it' < *ta-* IND.1/2.A + *t-* 1.A + *huras* 'find' + -Ø PERF

Loss of *h* also occurs when it precedes either a single word-final consonant or a two-consonant cluster. Thus,

$$h \rightarrow \emptyset \ / \ _ \ C\#, CC$$

ahiht 'be fat': *taatahit* 'I am fat' < *ta-* IND.1/2.A + *t-* 1.A+ *ahiht* 'be fat' + -Ø PERF; *ahitki* 'fat' < *ahiht* 'be fat' + *-kis* DIM; *ahitkuhku* 'bacon' < *ahiht* 'be fat' + *kuhkus* 'swine'

tiiwacpiitit 'he finally sat down' < *ti-* IND + Ø- 3.A + *awahc-* 'finally' + *wiitik* 'sit down' + -Ø PERF

Rule 17. Sibilant Hardening Rule. The sibilant *s* becomes *c* when it occurs after any consonant. Thus,

$$s \rightarrow c \ / \ C \ _$$

sa 'be lying': *tihcawaa* 'they are lying (there)' < *ti-* IND + *ir-* 3.PL.A + *sa* 'be lying' + *-waa* DIST + -Ø PERF

sakuriwihc (*uur* . . .) 'sun to come out': *tuhcakuriwihcuʔ* 'the sun is coming out' < *ti-* IND + Ø- 1.A + *uur-* PREV + *sakuriwihc* + *-:hus* IMPF

Rule 18. Sibilant Loss Rule. When *c* immediately follows the two-consonant cluster *ks*, the *s* of the cluster is dropped. Thus,

$$s \rightarrow \emptyset \ / \ k \ _ \ c$$

sucukcii 'be energetic!' < *i-* CONT.1/2A + *s-* 2.A + *ut-* PREV + *i-* SEQ + *uks-* JUSS + *ciir* 'be energetic' + -Ø (*-uk* 'be')

Rule 19. Alveolar Dissimilation Rule. A common underlying sequence across morpheme boundaries is *t* + *t*. In such a sequence the preceding *t* changes to *c*, as in the following examples:

tactaʔuut 'I stole it' < *ta-* IND.1/2.A + *t-* 1.A + *taʔuut* 'steal' + -Ø PERF

kiiwaʔuctaakaa 'white sage' < *kiiwaʔut* 'sage' + *taakaar* 'be white'

The less common underlying sequence *t* + *c* metathesizes, resulting in *ct*, as in:

tactat 'I shot it' < *ta-* IND.1/2A + *t-* 1.A + *cak* 'shoot' + -Ø PERF

wiituctactaaka 'the plant is sprouting' < *wii-* 'now' + *ti-* IND + Ø- 3.A + *ut-* PREV + *tat-* 'stalk; plant' + *caaka* 'be protruding' + -Ø PERF

Rule 20. Consonant Degemination Rule. When two identical consonants come together, one is lost, resulting in a single (short) consonant. This rule applies only after the consonantal changes specified by previous rules. Hence it operates for identical sequences of only two phonemes, *r* and *k*. Thus,

$$C_iC_i \rightarrow C_i$$

Examples of this rule are the following:

tiraraahuʔ 'they are ruined' < *ti-* IND + *Ø-* 3.A + *raar-* PL.INDV.A + *raahur* 'be ruined' + *-Ø* PERF

tuurariiʔi 'it is a mark' < *ti-* IND + *Ø-* 3.A + *uur-* PREV + *rariir* 'be marked' + *-Ø* PERF

tuutakaksawaa 'he called them' < *ti-* IND + *Ø-* 3.A + *ut-* BEN + *ak-* 3.P.AN.P + *kaksaa-* 'call' + *-waa* DIST + *-Ø* PERF

Rule 21. r-Stopping Rule. The sonorant *r* becomes the voiceless stop *t* when occurring after an obstruent. Thus,

$$r \rightarrow t \ / \ \text{Obs.} \ _$$

*kiri**k**taahkaa* 'wear eyeglasses' < *kiri**k**-* 'eye' + *raar-* PL + *kaa* 'be inside'

*suk**st**iisaapu* 'stake it!' < *i-* CONT.1/2.A + *s-* 2.A + *uks-* JUSS + *riisaapuh* + 'stake, stake down' + *-Ø* PERF

*iraa**c**tiʔ* 'my brother (female speaker)' < *i-* 1.POSS.A + *raa**c**-* 'brother' + *-rii* 1.POSS.B

Note that *t* is not included among the obstruents to which this rule applies since *t* has already been changed to *h* when it precedes *r* (see rule above); that is, the combination *t* + *r* changes to *hr* (which then reduces to *h*).

Rule 22. Labial Glide Loss. Although the details of this process are more complex than presented here, in verb derivation a *w* in a preceding syllable is generally lost when the following morpheme begins in *w*. Thus,

$$w \rightarrow \text{Ø} \ / \ \text{V} \ _ \ \text{V} + w\text{V(V)} \ \ldots$$

*a**w**iwiiʔa* (*ir* . . .) 'to bend over': *taaʔiwiiʔaʔ* 'he bent over' < *ti-* IND + *Ø-* 3.A + *a-* PREV + *a**w**iwiiʔa* 'bend over' + *-Ø* PERF

*kata**w**iwarik* 'swallow': *tatkataʔiwarit* 'I swallowed it' < *ta-* IND.1/2.A + *t-* 1.A + *kata**w**iwarik* 'swallow' + *-Ø* PERF

Rule 23. Final *r* Loss. Word-final *r* is always dropped. Thus,

$$r \rightarrow \text{Ø} \ / \ _ \ \#$$

wiitatuutaa 'I have done it' < *wii-* 'now' + *ta-* IND.1/2.A + *t-* 1.A + *ut-* PREV + *aar* + 'do' + *-Ø* PERF; cf. *wiitatuutaariʔ* 'I am doing it'

Rule 24. Variants of *c*. The phoneme *c* has two phonetic variants. One is [ts], which occurs before consonants and in word-final position; the other is [č], which occurs before vowels. Thus,

$$c \rightarrow \begin{bmatrix} [\text{ts}] \ / \ _ \ \text{C}, \# \\ [\check{\text{c}}] \ / \ _ \ \text{V} \end{bmatrix}$$

Examples are as follows:

capaat [čəpaat] ‘woman’

custit [čustɪt] ‘old woman’

ckuraaʔuʔ [tskuraaʔuʔ] ‘female doctor; nurse’

cpasaasi [tspəsaasi] ‘Osage Woman’

askuc [əskuts] ‘prairie dog’

tikiʔic [tikiʔits] ‘it’s watery’